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SPRING AI

GENERATIVE ARTIFICIAL INTELLIGENCE CON JAVA

Simone Scannapieco

Corso avanzato per Venis S.p.A, Venezia, Italia

Novembre 2025

Note

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PARTE 1 - APPROCCIO TEXT TO VECTOR STORE

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⚠️ Approccio *naive* (informazione testuale estratta tramite GENAI-assisted web scraping/summarization)

- 1 Iscrizione al portale HuggingFace e creazione *access token*
- 2 Verifica e modifica variabili di ambiente
- 3 Creazione configurazione ambiente Docker Qdrant
- 4 Modifica dipendenze di progetto
- 5 Modifica configurazione e profilo applicativo
- 6 Creazione *prompt templates* per strategia RAG
- 7 Configurazione *vector store* per Ollama e Gemini *embeddings*
- 8 Creazione *component* per popolamento *vector store* (dati Venis e CV)
- 9 Creazione interfaccia e implementazione del servizio
- 10 Modifica controllore MVC
- 11 Test delle funzionalità con Postman/Insomnia

Note

[illegible]

Note

[illegible]

**Hugging Face**

Q Search models, datasets, users...

Models

Datasets

Spaces

Community

Docs

Enterprise

Pricing

Log In

Sign Up



Join Hugging Face

Join the community of machine learners!

Email Address

Hint: Use your organization email to easily find and join your company/team org.

Password

- ✓ Must contain at least 8 characters
- ✓ Must contain uppercase, lowercase letters, and numbers
- ✓ If less than 12 characters, must contain a special character

Next

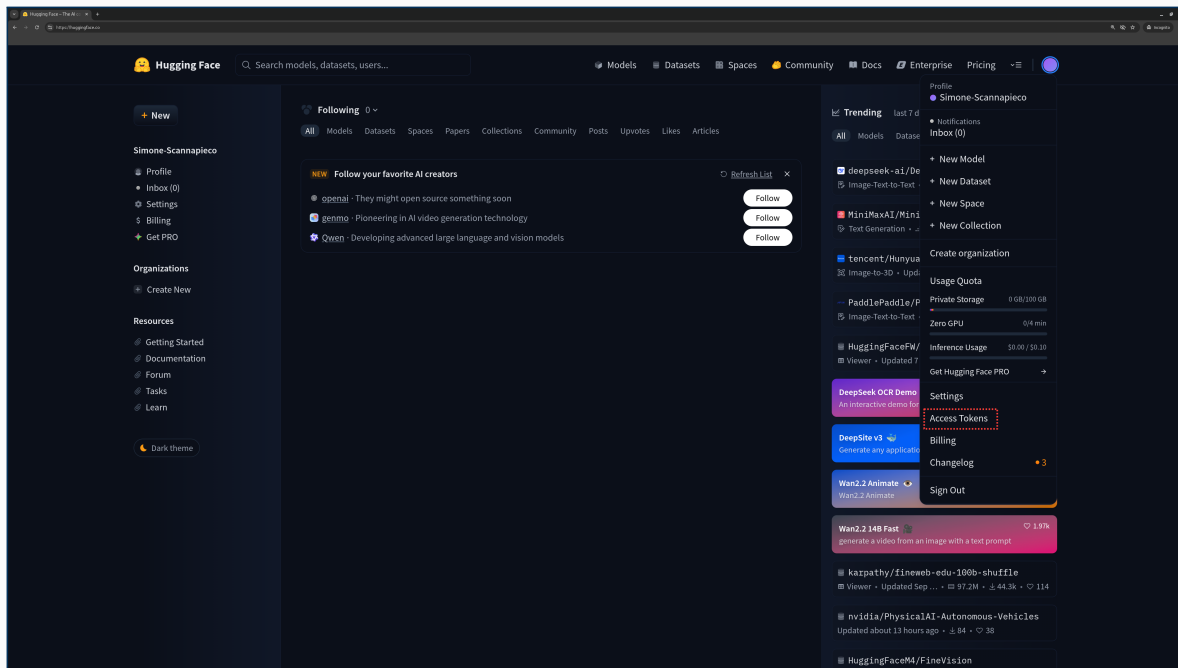
Already have an account? [Log in](#)

SSO is available for [Team](#) & [Enterprise](#) accounts.

Note

[illegible]

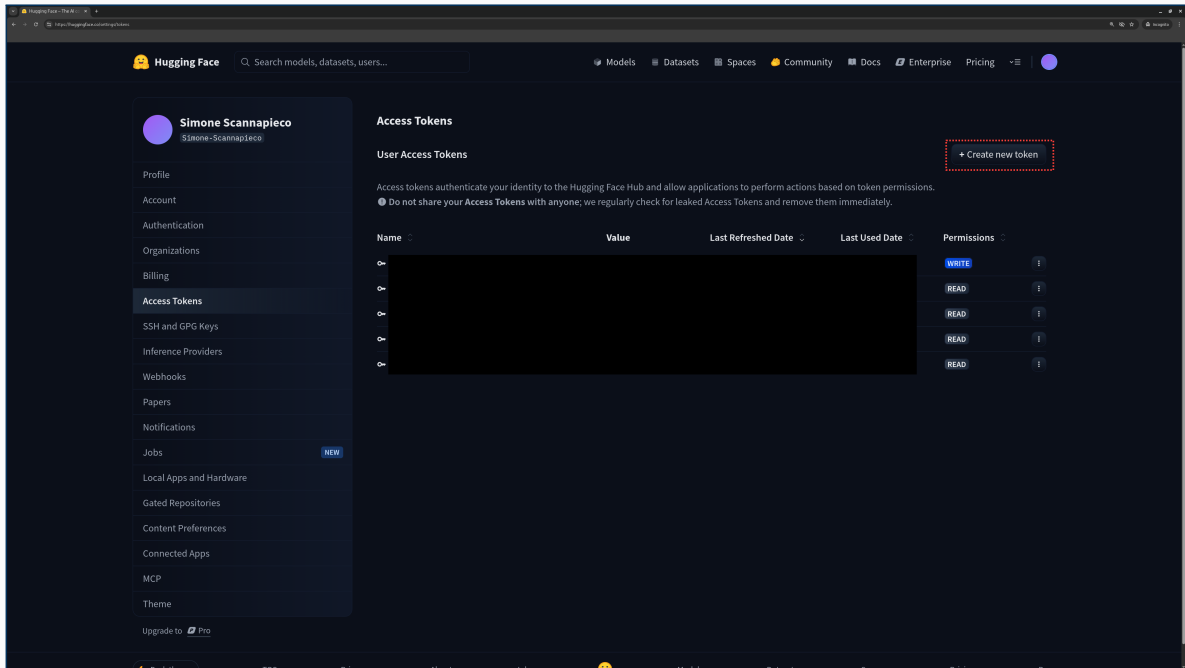
3



Note

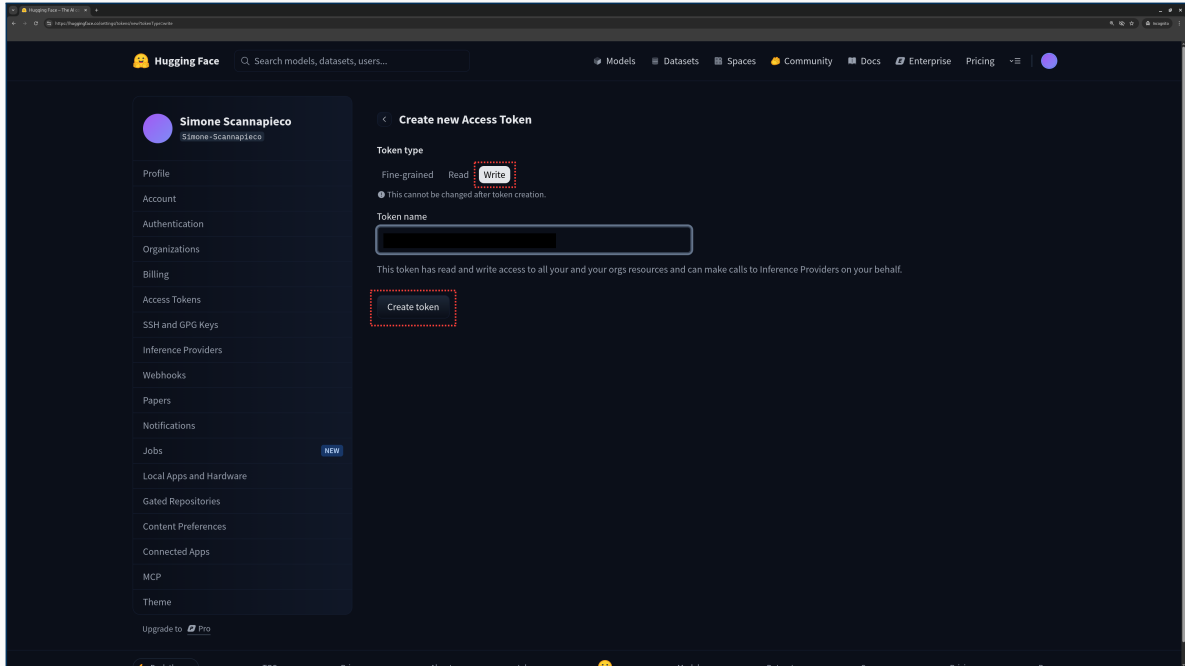
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4



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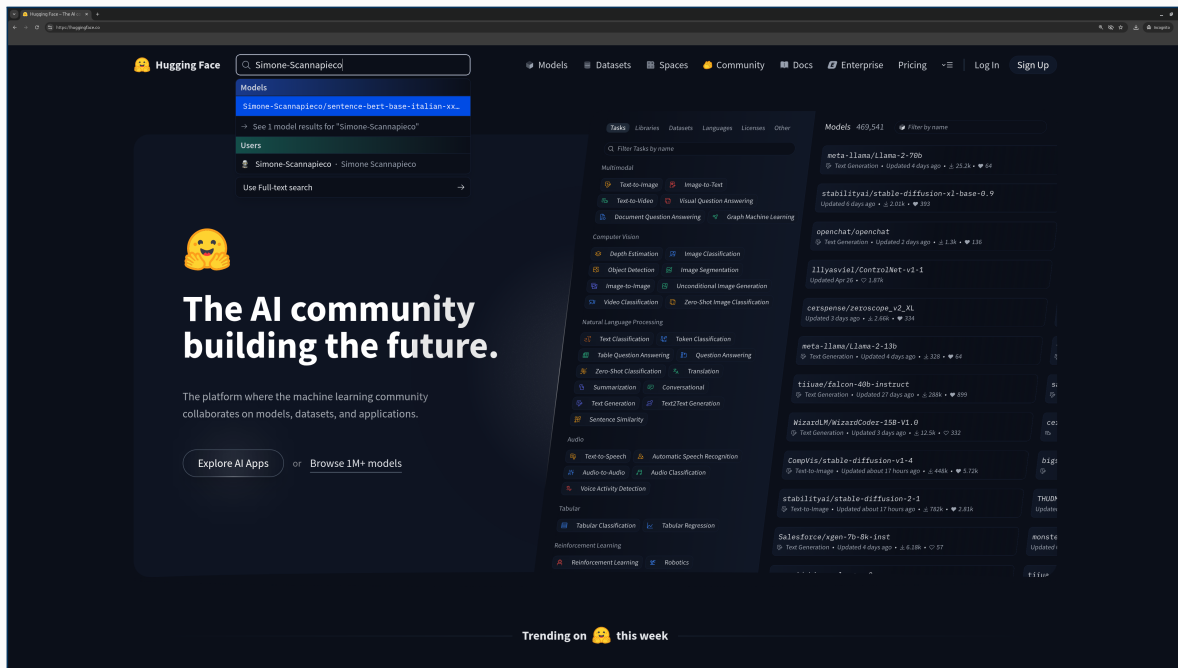
5



Note

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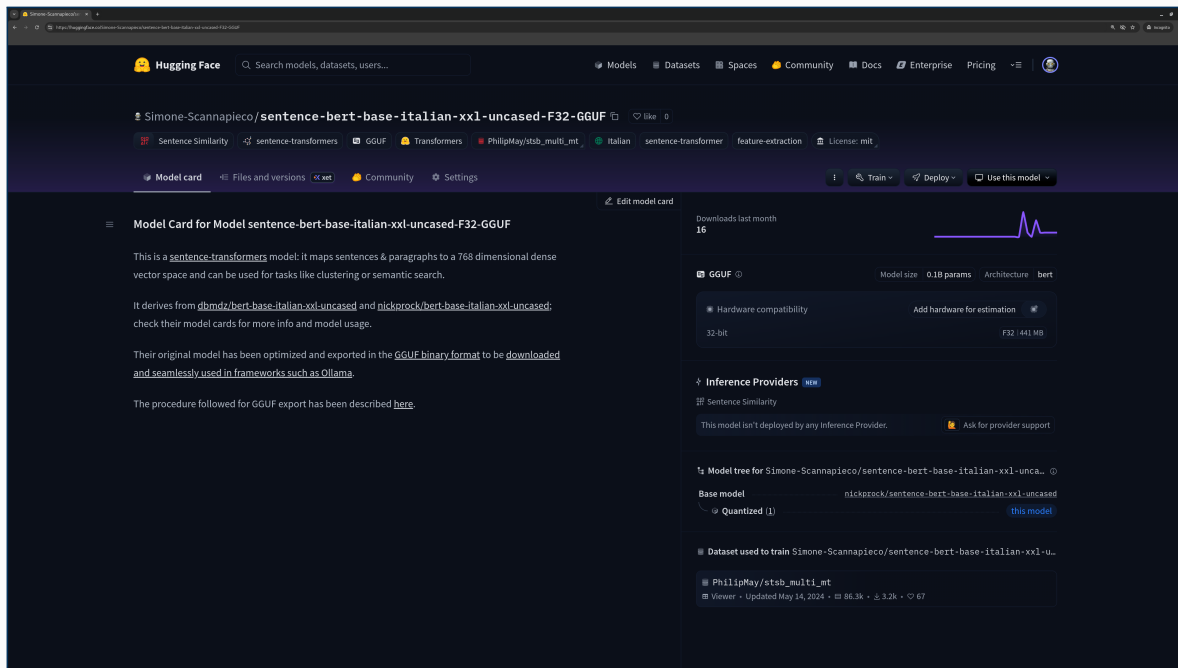
6 Cercare il modello di *embedding*, leggendone la *card*



Note

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6 Cercare il modello di *embedding*, leggendone la *card*



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```
{
  // Use IntelliSense to learn about possible attributes.
  // Hover to view descriptions of existing attributes.
  // For more information, visit: https://go.microsoft.com/fwlink/?linkid=830387
  "version": "0.2.0",
  "configurations": [
    {
      "type": "java",
      "name": "Launch Current File",
      "request": "launch",
      "mainClass": "${file}"
    },
    {
      "type": "java",
      "name": "DemoApplication",
      "request": "launch",
      "mainClass": "it.venis.ai.spring.demo.DemoApplication",
      "projectName": "demo",
      "env": {
        "GOOGLE_AI_API_KEY": "...",
        "HUGGING_FACE_HUB_TOKEN": "..."
      }
    }
  ]
}
```

Note

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```
{
  "java.compile.nullAnalysis.mode": "disabled",
  "java.configuration.updateBuildConfiguration": "interactive",
  "java.test.config": {
    "env": {
      "GOOGLE_AI_API_KEY": "...",
      "HUGGING_FACE_HUB_TOKEN": "..."
    }
  }
}
```

[illegible]

```
services:
  ...
  spring-ai-vector-store:
    image: qdrant/qdrant:${QDRANT_VERSION:-latest}
    hostname: spring-ai-vector-store
    container_name: spring_ai_vector_store
    ports:
      - ${QDRANT_HTTP_PORT:-6333}:6333
      - ${QDRANT_GRPC_PORT:-6334}:6334
    volumes:
      - spring_ai_vector_store:/qdrant/storage
    restart: unless-stopped

volumes:
  ...
  spring_ai_vector_store:
    name: spring_ai_vector_store
```

[illegible]

```
...
# qdrant configuration
QDRANT_VERSION=v1.13.0                                # default: latest
QDRANT_HTTP_PORT=6333                                  # default: 6333
QDRANT_GRPC_PORT=6334                                  # default: 6334
```

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```
...
<dependency>
    <groupId>org.springframework.ai</groupId>
    <artifactId>spring-ai-rag</artifactId>
</dependency>
<dependency>
    <groupId>org.springframework.ai</groupId>
    <artifactId>spring-ai-advisors-vector-store</artifactId>
</dependency>
<dependency>
    <groupId>org.springframework.ai</groupId>
    <artifactId>spring-ai-starter-vector-store-qdrant</artifactId>
</dependency>
...
```

Note

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, leaving small margins at the top and bottom. There are no vertical margin lines, and the paper is otherwise completely blank.


```
spring:
  ...
  profiles:
    active: rag-text-to-vector-store
  autoconfigure:
    exclude:
      - org.springframework.ai.vectorstore.qdrant.autoconfigure.QdrantVectorStoreAutoConfiguration
      # We must disable Vector Store auto-configuration because of two different EmbeddingModel beans
      # (OpenAI-Gemini and Ollama). The goal is to have two different vector collections, one for each
      # family of LLM.
  ai:
    ollama:
      ...
      embedding:
        model: hf.co/Simone-Scannapieco/sentence-bert-base-italian-xxl-uncased-F32-GGUF
    openai:
      ...
      embedding:
        options:
          model: gemini-embedding-001
  vectorstore:
    qdrant:
      initialize-schema: true
      host: 172.17.0.1
      port: 6334
  ...
```

Note

[illegible]

```
demo:
  rag:
    prompt:
      system:
        ita: classpath:templates/get-rag-data-system-ita-prompt.st
        eng: classpath:templates/get-rag-data-system-eng-prompt.st
      user:
        ita:
        eng:
    vectorstore:
      qdrant:
        collection-name:
          ollama: vector_store_ttvvs_ollama
          gemini: vector_store_ttvvs_gemini
```

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```
#!/bin/bash

volume_name=spring_ai_llm

docker volume rm $volume_name
```

```
#!/bin/bash

volume_name=spring_ai_vector_store

docker volume rm $volume_name
```

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```
Sei un assistente AI in grado di rispondere alle domande dell'utente solo in base
al contesto fornito dalla sezione DOCUMENTI.
Se la risposta non è presente nella sezione DOCUMENTI, informa l'utente di non
sapere la risposta.
```

You are an helpful assistant, answering questions based on the given context in the DOCUMENTS section and no prior knowledge.
If the answer is not in the DOCUMENTS section, then reply that you cannot answer to the question.

Note

```
package it.venis.ai.spring.demo.config;

...

@Configuration
public class RAGConfig {

    @Value("${spring.ai.vectorstore.qdrant.host:localhost}")
    private String qdrantHost;
    @Value("${spring.ai.vectorstore.qdrant.port:6334}")
    private Integer qdrantPort;
    @Value("${spring.ai.vectorstore.qdrant.use-tls:false}")
    private Boolean useTls;

    @Bean
    public QdrantClient qdrantClient() {
        QdrantGrpcClient.Builder grpcClientBuilder = QdrantGrpcClient.newBuilder(
            qdrantHost, qdrantPort, useTls);
        return new QdrantClient(grpcClientBuilder.build());
    }

    @Value("${demo.rag.vectorstore.qdrant.collection-name.gemini:vector_store_gemini}")
    private String qdrantCollectionNameGemini;
    @Value("${demo.rag.vectorstore.qdrant.collection-name.ollama:vector_store_ollama}")
    private String qdrantCollectionNameOllama;
    @Value("${spring.ai.vectorstore.qdrant.initialize-schema:false}")
    private Boolean qdrantInitializeSchema;

    ...
}
```

Note

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Configurazione Vector Store - II

```
...
@Bean
public VectorStore geminiVectorStore(QdrantClient qdrantClient, OpenAiEmbeddingModel geminiEmbeddingModel) {
    return QdrantVectorStore.builder(qdrantClient, geminiEmbeddingModel)
        .collectionName(qdrantCollectionNameGemini)
        .initializeSchema(qdrantInitializeSchema)
        .build();
}

@Bean
public VectorStore ollamaVectorStore(QdrantClient qdrantClient, OllamaEmbeddingModel ollamaEmbeddingModel) {
    return QdrantVectorStore.builder(qdrantClient, ollamaEmbeddingModel)
        .collectionName(qdrantCollectionNameOllama)
        .initializeSchema(qdrantInitializeSchema)
        .build();
}
}
```

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Componente popolamento Qdrant - I

```
package it.venis.ai.spring.demo.rag;

...

@Component
@Profile("rag-text-to-vector-store")
public class TextDataLoader {

    private final VectorStore geminiVectorStore;
    private final VectorStore ollamaVectorStore;

    public TextDataLoader(@Qualifier("geminiVectorStore") VectorStore geminiVectorStore,
        @Qualifier("ollamaVectorStore") VectorStore ollamaVectorStore) {
        this.geminiVectorStore = geminiVectorStore;
        this.ollamaVectorStore = ollamaVectorStore;
    }

    @PostConstruct
    public void loadVenisInfoIntoVectorStore() {
        List<String> venisInfo = List.of(...);
        SearchRequest searchRequest = SearchRequest.builder()
            .query("Check")
            .similarityThresholdAll()
            .build();
        List<Document> similarDocs = geminiVectorStore.similaritySearch(searchRequest);
        if (similarDocs.size() == 0) {
            List<Document> documents =
                venisInfo.stream().map(Document::new).collect(Collectors.toList());
            this.geminiVectorStore.add(documents);
        }
    }
}
```

```
...
@PostConstruct
public void loadSSCVInfoIntoVectorStore() {
    List<String> ssCVInfo = List.of(...);
    SearchRequest searchRequest = SearchRequest.builder()
        .query("Check")
        .similarityThresholdAll()
        .build();
    List<Document> similarDocs = ollamaVectorStore.similaritySearch(searchRequest);
    if (similarDocs.size() == 0) {
        List<Document> documents = ssCVInfo.stream().map(Document::new).collect(Collectors.toList());
        this.ollamaVectorStore.add(documents);
    }
}
}
```

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```
package it.venis.ai.spring.demo.services;

import it.venis.ai.spring.demo.model.Answer;
import it.venis.ai.spring.demo.model.QuestionRequest;

public interface RAGService {

    public Answer getGeminiRAGAnswer(QuestionRequest request);

    public Answer getOllamaRAGAnswer(QuestionRequest request);

}
```

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```
package it.venis.ai.spring.demo.services;

...

@Service
@Configuration
public class RAGServiceImpl implements RAGService {

    private final ChatClient geminiChatClient;
    private final ChatClient ollamaChatClient;
    private final ChatClient ollamaMemoryChatClient;
    private VectorStore geminiVectorStore;
    private VectorStore ollamaVectorStore;

    public RAGServiceImpl(
        @Qualifier("geminiChatClient") ChatClient geminiChatClient,
        @Qualifier("ollamaChatClient") ChatClient ollamaChatClient,
        @Qualifier("ollamaMemoryChatClient") ChatClient ollamaMemoryChatClient,
        @Qualifier("geminiVectorStore") VectorStore geminiVectorStore,
        @Qualifier("ollamaVectorStore") VectorStore ollamaVectorStore) {

        this.geminiChatClient = geminiChatClient;
        this.ollamaChatClient = ollamaChatClient;
        this.ollamaMemoryChatClient = ollamaMemoryChatClient;
        this.geminiVectorStore = geminiVectorStore;
        this.ollamaVectorStore = ollamaVectorStore;
    }

    ...
}
```

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```
...
@Value("${demo.rag.prompt.system.eng}")
private Resource ragDataSystemEngPrompt;

@Override
public Answer getGeminiRAGAnswer(QuestionRequest request) {

    SearchRequest searchRequest = SearchRequest.builder()
        .query(request.body().question())
        .topK(4)
        .similarityThreshold(.2)
        .build();

    List<Document> similarDocs = geminiVectorStore.similaritySearch(searchRequest);

    String similarDocsString = similarDocs.stream()
        .map(Document::getText)
        .collect(Collectors.joining(System.lineSeparator()));

    return new Answer(this.geminiChatClient.prompt()
        // .advisors(advisorSpec -> advisorSpec.param(ChatMemory.CONVERSATION_ID,
        // request.username()))
        .system(s -> s.text(this.ragDataSystemEngPrompt)
            .params(Map.of("document1", similarDocsString)))
        .user(request.body().question())
        .templateRenderer(StTemplateRenderer.builder().startDelimiterToken('<')
            .endDelimiterToken('>')
            .build())
        .call()
        .content());
}
...
```

Note

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```

...
@Value("${demo.rag.prompt.system.ita}")
private Resource ragDataSystemItaPrompt;

@Override
public Answer getOllamaRAGAnswer(QuestionRequest request) {

    SearchRequest searchRequest = SearchRequest.builder()
        .query(request.body().question())
        .topK(4)
        .similarityThreshold(.3)
        .build();

    List<Document> similarDocs = ollamaVectorStore.similaritySearch(searchRequest);

    String similarDocsString = similarDocs.stream()
        .map(Document::getText)
        .collect(Collectors.joining(System.lineSeparator()));

    return new Answer(this.ollamaMemoryChatClient.prompt()
        .advisors(advisorSpec -> advisorSpec.param(ChatMemory.CONVERSATION_ID, request.username()))
        .system(s -> s.text(this.ragDataSystemItaPrompt)
            .params(Map.of("document1", similarDocsString)))
        .user(request.body().question())
        .templateRenderer(StTemplateRenderer.builder().startDelimiterToken('<')
            .endDelimiterToken('>')
            .build())
        .call()
        .content());
}
}

```

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```
package it.venis.ai.spring.demo.controllers;

...

@RestController
public class QuestionController {

    private final QuestionService service;
    private final RAGService ragService;

    public QuestionController(QuestionService service, RAGService ragService) {
        this.service = service;
        this.ragService = ragService;
    }

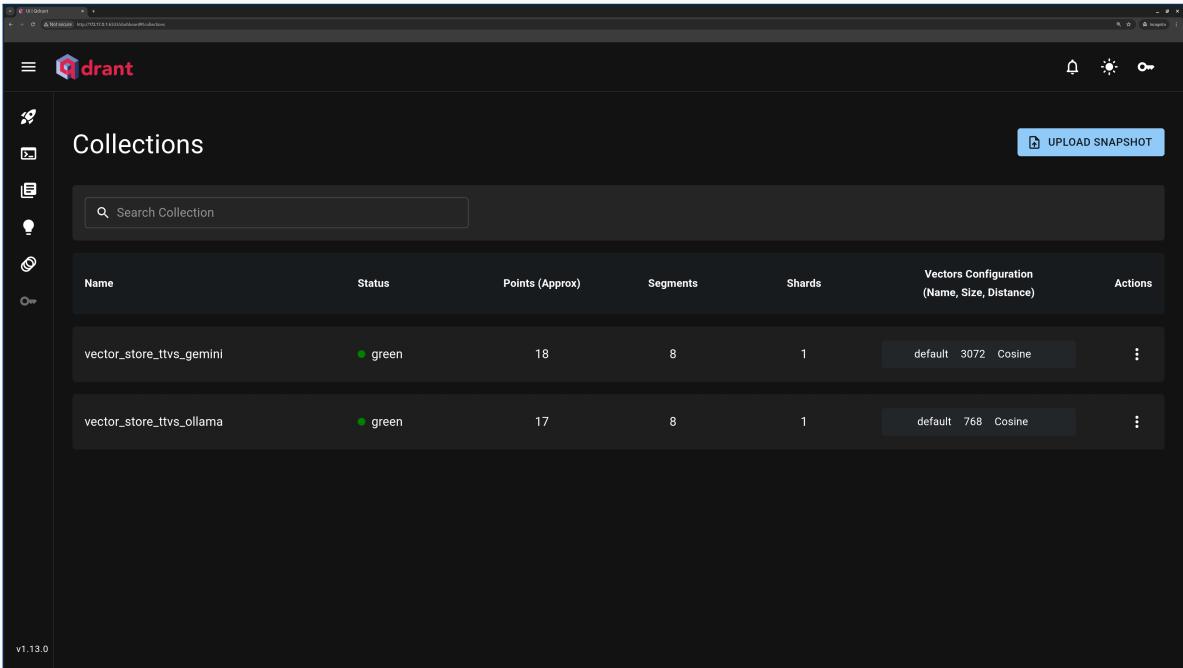
    ...

    @PostMapping("/gemini/ask/rag")
    public Answer getGeminiRAGAnswer(@RequestBody QuestionRequest request) {
        return this.ragService.getGeminiRAGAnswer(request);
    }

    @PostMapping("/ollama/ask/rag")
    public Answer getOllamaRAGAnswer(@RequestBody QuestionRequest request) {
        return this.ragService.getOllamaRAGAnswer(request);
    }
}
```

[illegible]


 Verificare la *dashboard* Qdrant (<http://172.17.0.1:6333/dashboard>)



Note

A series of horizontal lines for taking notes.

24 / 24

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